

CUSTOMER CHURN PREDICTION USING MACHINE LEARNING

PROJECT REPORT

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ABSTRACT

Customer churn is a major concern in the banking sector because losing existing customers can affect business growth and profitability. This project focuses on analyzing customer banking data and developing a machine learning model to predict customer churn.

The dataset was analyzed using Exploratory Data Analysis (EDA) to identify patterns and factors associated with customer churn. Important attributes such as age, gender, geography, credit score, balance, number of products, tenure, estimated salary, and active membership were considered. Two machine learning algorithms, Random Forest and Logistic Regression, were implemented and evaluated.

The Random Forest model achieved an accuracy of 86.35% and an AUC score of 0.85, outperforming Logistic Regression, which achieved an accuracy of 80.45% and an AUC score of 0.77. The analysis identified age as the most important feature in the Random Forest model. Customers aged 51–60 showed the highest churn rate of 56.21%, while Germany had the highest country-level churn rate at 32.44%.

The results demonstrate that machine learning can help identify customer churn patterns and support banks in developing targeted customer-retention strategies.

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1. INTRODUCTION

Customer churn refers to the situation in which a customer stops using the services provided by a company. In the banking sector, customer churn is an important business concern because retaining existing customers is valuable for maintaining long-term relationships and business growth.

With the increasing availability of customer data, machine learning can be used to analyze customer behavior and identify patterns associated with churn. By studying demographic, financial, and account-related information, organizations can identify customers who may be at a higher risk of leaving.

This project focuses on customer churn prediction using machine learning. A banking customer dataset is analyzed to understand the factors associated with churn. Exploratory Data Analysis is performed using customer attributes such as age, gender, geography, credit score, balance, number of products, tenure, estimated salary, and active membership.

Two machine learning algorithms, Random Forest Classifier and Logistic Regression, are implemented to predict whether a customer is likely to churn. Their performance is evaluated using accuracy, confusion matrix, classification report, and Area Under the ROC Curve (AUC).

The Random Forest model achieved an accuracy of 86.35% and an AUC score of 0.85, performing better than Logistic Regression.

2. PROBLEM STATEMENT

Customer churn can negatively affect the growth and profitability of banking institutions. Identifying customers who are likely to leave can help banks take preventive measures and improve customer retention.

The problem addressed in this project is to analyze banking customer data and develop a machine learning model that can predict whether a customer is likely to churn while identifying the major factors associated with customer churn.

3. OBJECTIVES

The main objectives of this project are:

- * To analyze banking customer data and understand customer churn patterns.
- * To identify factors associated with customer churn.
- * To perform data preprocessing and prepare the dataset for machine learning.
- * To perform Exploratory Data Analysis on customer information.
- * To develop a customer churn prediction model using Random Forest.
- * To implement Logistic Regression for comparison.
- * To evaluate the models using accuracy, confusion matrix, classification report, and AUC.
- * To identify important features influencing churn prediction.
- * To provide useful insights that can support customer-retention strategies.

4. DATASET DESCRIPTION

The project uses a banking customer dataset containing demographic, financial, and account-related information.

The major attributes used in the analysis include:

Feature	Description
RowNumber	Unique row identifier
CustomerID	Unique customer identification number
Surname	Customer surname
Creditscore	Customer creditscore
Geography	Country of the customer
Gender	Gender of the customer
Age	Age of the customer
Tenure	Number of years the customer has been with the bank
Balance	Customer's bank account balance
NumOfProducts	Number of bank products used
HasCrCard	Indicates whether the customer has a credit card
IsActiveMember	Indicates whether the customer is an active member
EstimatedSalary	Estimated customer salary
Exited	Target variable indicating whether the customer churned

The target variable is Exited.

* 0: Customer stayed with the bank

* 1: Customer churned

The dataset was divided into training and testing sets using an 80:20 split.

* Training data: 8,011 records

* Testing data: 2,003 records

5. METHODOLOGY

The project follows a systematic machine learning workflow.

5.1 Data Collection

The banking customer dataset was obtained and loaded into the Google Colab environment using Python.

5.2 Data Preprocessing

The dataset was inspected and prepared for analysis. Categorical variables were encoded into numerical values so that they could be used by machine learning algorithms.

5.3 Exploratory Data Analysis

Customer churn rates were calculated across different demographic and account-related groups. Visualizations were also created to identify important patterns.

5.4 Feature Preparation

Relevant features were separated from the target variable Exited. Unnecessary identification columns were removed before model training.

5.5 Train-Test Split

The dataset was divided into training and testing data using an 80:20 split.

5.6 Model Development

Two machine learning algorithms were used:

- * Random Forest Classifier
- * Logistic Regression

5.7 Model Evaluation

The models were evaluated using:

- * Accuracy
- * Confusion Matrix
- * Precision
- * Recall
- * F1-score
- * ROC Curve
- * AUC Score

5.8 Model Selection

The models were compared based on accuracy and AUC. Random Forest achieved better performance and was selected as the better-performing model for this project.

6. DATA PREPROCESSING

Data preprocessing is an important stage in machine learning because raw data may contain unnecessary attributes or categorical values that cannot be directly used by the model.

In this project, the following preprocessing steps were performed:

6.1 Categorical Encoding

The Geography and Gender columns were converted into numerical values using Label Encoding.

6.2 Removal of Unnecessary Columns

Columns such as RowNumber, CustomerId, and Surname were removed because they are identifiers and do not provide useful predictive information.

Additional analysis-only grouping columns such as AgeGroup and SalaryGroup were also excluded from the model training data.

6.3 Feature and Target Separation

The Exited column was selected as the target variable, while the remaining relevant columns were used as input features.

6.4 Train-Test Split

The dataset was divided into:

- * 80% training data
- * 20% testing data

The training dataset contained 8,011 records, while the testing dataset contained 2,003 records.

7. EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) was performed to understand customer behavior and identify patterns associated with customer churn. Different demographic, financial, and account-related attributes were analyzed.

7.1 Overall Churn Rate

- The overall customer churn rate was calculated using the Exited variable.
- The overall churn rate was 20.37%.
- This means that approximately one-fifth of the customers in the dataset had exited the bank.

7.2 Churn Rate by Gender

- The churn rate was analyzed based on gender.
- Female customers had a churn rate of 25.07%, while male customers had a churn rate of 16.40%.
- The results show that female customers had a higher observed churn rate than male customers in this dataset.

7.3 Churn Rate by Geography

- The churn rate was analyzed for customers from France, Germany, and Spain.
- France had a churn rate of 16.15%.
- Germany had the highest churn rate at 32.44%.
- Spain had a churn rate of 16.67%.

Therefore, Germany showed the highest observed customer churn among the three countries.

7.4 Churn Rate by Age Group

Customers were divided into different age groups to understand the relationship between age and churn.

- Customers aged 18–30 had the lowest churn rate of 7.50%.
- Customers aged 31–40 had a churn rate of 12.08%.
- Customers aged 41–50 had a churn rate of 33.96%.
- Customers aged 51–60 had the highest churn rate of 56.21%.
- Customers aged 60 and above had a churn rate of 24.78%.

The results show a substantial increase in observed churn among customers in the 41–60 age range, with the 51–60 group having the highest churn rate.

7.5 Churn Rate by Tenure

The relationship between customer tenure and churn was also analyzed.

- Customers with 0 years of tenure had an observed churn rate of approximately 23.00%.
- The churn rate for customers with 1 year of tenure was approximately 22.40%.
- Customers with 2 years of tenure had a churn rate of approximately 19.18%, while customers with 3 years had approximately 21.11%.
- The churn rate remained relatively similar for most other tenure groups. Customers with 7 years of tenure had the lowest observed churn rate at approximately 17.21%.

Overall, tenure did not show as strong a difference in churn as age or geography.

7.6 Churn Rate by Estimated Salary

Customers were divided into four groups based on their estimated salary.

- The low-salary group had a churn rate of 20.00%.
- The medium-low salary group had a churn rate of 19.80%.
- The medium-high salary group had a churn rate of 20.12%.
- The high-salary group had the highest churn rate among the salary groups at 21.56%.

The differences between the groups were relatively small, suggesting that salary group alone was not a strong differentiating factor in the observed churn rates.

7.7 Churn Rate by Active Membership

Customer activity status was analyzed to determine whether active membership was associated with churn.

- Inactive members had an observed churn rate of approximately 26.85%, while active members had a churn rate of approximately 14.26%.
- This indicates that inactive members had a considerably higher observed churn rate than active members.

Therefore, customer engagement and activity may be useful factors to consider when developing customer-retention strategies.

7.8 Summary of Exploratory Data Analysis

The EDA revealed several important patterns in customer churn. Age showed a particularly strong difference, with customers aged 51–60 having the highest observed churn rate. Germany also had a considerably higher churn rate compared with France and Spain. Female customers and inactive members showed higher observed churn rates than their respective comparison groups.

These findings helped identify potentially important variables for the subsequent machine learning analysis.

8. MACHINE LEARNING MODELS

Two machine learning algorithms were implemented in this project.

8.1 Random Forest Classifier

Random Forest is an ensemble machine learning algorithm that combines multiple decision trees to make predictions.

It was selected because it can capture complex relationships between customer attributes and churn. It also provides feature-importance information that helps identify the variables contributing to model predictions.

The Random Forest model was trained using the training dataset and evaluated using the testing dataset.

8.2 Logistic Regression

Logistic Regression is a classification algorithm commonly used for predicting binary outcomes.

In this project, Logistic Regression was implemented as a baseline model and compared with Random Forest to determine which model performed better.

9. MODEL EVALUATION

The machine learning models were evaluated using several performance metrics.

9.1 Model Accuracy

- The Random Forest model achieved an accuracy of 86.35%.
- The Logistic Regression model achieved an accuracy of 80.45%.
- Therefore, Random Forest achieved higher prediction accuracy.

9.2 AUC Comparison

- The Random Forest model achieved an AUC score of 0.85.
- The Logistic Regression model achieved an AUC score of 0.77.

The higher AUC value of Random Forest indicates better ability to distinguish between customers who churn and customers who stay.

9.3 Confusion Matrix

The confusion matrix of the Random Forest model showed 1,538 correctly predicted customers who stayed and 189 correctly predicted customers who churned.

The model incorrectly classified 55 customers who stayed as churned and 218 customers who actually churned as customers who would stay.

9.4 Classification Report

- The classification report was used to evaluate precision, recall, F1-score, and support.
- For customers who stayed, the model achieved a precision of 0.88 and recall of 0.97.
- For customers who churned, the model achieved a precision of 0.77 and recall of 0.46.

The churn-class recall of 0.46 indicates that there is room for improvement in identifying all customers who actually churn.

9.5 ROC Curve and AUC

The ROC curve was used to evaluate the ability of the model to distinguish between customers who churn and customers who stay.

The Random Forest model achieved an AUC score of 0.85, indicating good discriminatory performance.

10. FEATURE IMPORTANCE

Feature importance was obtained from the Random Forest model to identify which variables contributed most to the model's predictions.

- Age was the most important feature, with an importance value of approximately 0.2384.
- Estimated Salary had an importance of approximately 0.1469.
- Credit Score had an importance of approximately 0.1459.

- Balance had an importance of approximately 0.1394.
- Number of Products had an importance of approximately 0.1329.
- Tenure had an importance of approximately 0.0823.
- Is Active Member had an importance of approximately 0.0386.
- Geography had an importance of approximately 0.0381.
- Gender had an importance of approximately 0.0183.
- Has Credit Card had an importance of approximately 0.0119
- Year had an importance of 0.0000.

Therefore, Age was identified as the top feature influencing the Random Forest model's predictions.

It is important to note that feature importance indicates how useful a feature was to the model's predictions. It does not by itself prove that a feature directly causes customer churn.

11. RESULTS AND KEY FINDINGS

The major findings of the project are as follows.

- The overall customer churn rate was 20.37%.
- Female customers showed a higher observed churn rate of 25.07% compared with male customers at 16.40%.
- Germany had the highest observed country-level churn rate at 32.44%.
- Customers aged 51–60 had the highest observed age-group churn rate at 56.21%.
- Customers aged 18–30 had the lowest observed age-group churn rate at 7.50%.

Inactive members had a higher observed churn rate of approximately 26.85% compared with active members at approximately 14.26%.

- Salary groups showed relatively small differences in churn rates.
- Age was the most important feature in the Random Forest model.
- Random Forest achieved 86.35% accuracy and an AUC score of 0.85.

- Logistic Regression achieved 80.45% accuracy and an AUC score of 0.77.

Random Forest therefore performed better than Logistic Regression based on accuracy and AUC.

The Random Forest model's recall for the churn class was 46%, indicating that improving the identification of actual churners could be an important area for future improvement.

12. CONCLUSION

The project successfully analyzed banking customer data and developed machine learning models for customer churn prediction.

Exploratory Data Analysis identified meaningful differences in churn across age groups, gender, geography, and membership activity. Customers aged 51–60 showed the highest observed churn rate, while Germany had the highest observed churn among the countries in the dataset.

Two machine learning models were developed and compared. Random Forest achieved an accuracy of 86.35% and an AUC score of 0.85, outperforming Logistic Regression, which achieved 80.45% accuracy and an AUC score of 0.77.

The feature-importance analysis identified Age as the most important feature used by the Random Forest model.

The results demonstrate that machine learning can be useful for identifying customer churn patterns and supporting data-driven customer-retention strategies.

However, the model does not guarantee that an individual customer will churn. Its predictions should be used as decision-support information along with appropriate business analysis.

13. FUTURE SCOPE

The project can be further improved in several ways.

Future work can include experimenting with additional machine learning algorithms such as XGBoost, Gradient Boosting, and Support Vector Machines.

Hyperparameter tuning can be performed to improve the performance of the Random Forest model.

The relatively low recall for the churn class can be addressed using class-weighting techniques or resampling methods.

The trained model can be developed into an interactive web application that allows users to enter customer details and obtain a churn prediction.

The model can also be deployed on a cloud platform for real-world use.

Customer-specific explanations can be added so that banks can understand why a particular customer has been classified as having a higher churn risk.

The model can be continuously retrained using updated customer data to maintain its performance over time.

The prediction system could also be integrated with customer relationship management systems to support targeted customer-retention strategies.

14. REFERENCES

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